INSTITUTION OF TECHNOLOGY

SCHOOLOF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE

COURSE TITLE:ADVANCED DATABASE

COURSE CODE:SOSC2042

PROJECT TITLE: INVENTOR CONTROL MANAGEMENT SYSTEM

GROUP 7

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WOLDIA ETHIOPIA

Table of Contents

Abstraction i
Introduction i
Background 1
Objective 2
Business Rules 4
ENTITIES 5
ENTITES AND ARTTIBUTIES 5
Relationships and Cardinality Ratios 8
CARDINALITY RATIO AND PARTICIPATION FOR ENTITIES 10
ER DAGRAMDIAGRAM 12
CONVERT ER DIAGRAM INTO TABLE 13
MAPPING RELATION 14
FUNCTIONAL DEPENDENCY 15
NORMALIZATION 16
SQL 17
Conclusion 27

Abstraction

An Inventor Management System (IMS) abstracts the processes of tracking and controlling the lifecycle of inventory items within an organization. It centralizes the collection of inventory data, including quantities, locations, and movements, allowing for efficient monitoring and reporting. The system typically integrates with other enterprise systems, streamlining procurement, stock management, and order fulfillment. Users interact with the IMS through a simple interface, focusing on key actions like inventory checks, reorder triggers, and audits. Overall, it enhances inventory accuracy, reduces waste, and ensures optimal stock levels.

Introduction

An Inventory Control Management System (ICMS) is a software solution designed to manage inventory levels, orders, sales, and deliveries. It helps businesses track inventory in real-time, ensuring optimal stock levels to meet customer demand while minimizing excess inventory

An Inventory Control Management System (ICMS) is a comprehensive software solution tailored to streamline and optimize the management of inventory across various industries. The primary goal of an ICMS is to ensure that businesses maintain the right balance of stock, thereby reducing costs and maximizing efficiency in operations

Background

With the rise of e-commerce and global supply chains, effective inventory management has become crucial for businesses of all sizes. Traditional manual methods are often inefficient and prone to errors, leading to lost sales, overstock, or stockouts. ICMS addresses these challenges by automating and streamlining inventory processes.

Objective

The primary objectives of an Inventory Control Management System include:

Real-time Inventory Trackin: Monitor stock levels continuously.

Order Management: Manage the procurement and fulfillment of inventory.

Data Analysis: Provide insights into inventory trends and performance.

- Cost Reduction**: Minimize holding costs and reduce wastage.

Scope

The scope of an ICMS encompasses:

Inventory Management: Tracking inventory in warehouses and retail locations.

Reporting: Generating reports on stock levels, sales, and forecasts.

integration: Working with other business systems like ERP and CRM.

User Access: Allowing various stakeholders (e.g., managers, sales staff) to access the system based on their roles.

Overview

An ICMS typically features:

Dashboard: A user-friendly interface displaying key metrics.

Inventory Database: A centralized database for all inventory items.

Alerts and Notification: Automated alerts for low stock levels or expirations.

Mobile Access: Enabling users to manage inventory on the go.

Business Rules

Key business rules for an ICMS may include:

Stock Reorder Levels: Setting minimum stock levels to trigger automatic replenishment.

Inventory Valuation Applying methods like FIFO (First In First Out) or LIFO (Last In, First Out) for valuation.

User Permissions: Defining access rights based on roles to maintain data integrity.

Audit Trails: Keeping logs of all inventory transactions for accountability.

ENTITIES

1. Product:

2. Supplier:

3. Warehouse:

4. Stock Level:

5. Inventory Transaction:

6. Purchase Order:

7. Sales Order:

ENTITES AND ARTTIBUTIES

1. Product

Attributes

Product ID (Primary Key)

Name

Description

SKU

Category

Price

Quantity in Stock

2. Supplier

Attributes

Supplier ID (Primary Key)

Name

Contact Name

Phone Number

Email

Address

3. Warehouse

Attributes:

Warehouse ID (Primary Key)

Name

Address

Capacity

4. Stock Level

Attributes:

Stock Level ID (Primary Key)

Product ID (Foreign Key)

Current Quantity

Minimum Threshold

Maximum Threshold

5. Inventory Transaction

Attributes:

Transaction ID (Primary Key)

Product ID (Foreign Key)

Quantity Changed

Transaction Type (e.g., addition, subtraction)

Date

6. Purchase Order

Attributes:

Purchase Order ID (Primary Key)

Supplier ID (Foreign Key)

Product ID (Foreign Key)

Order Quantity

Order Date

Status

7. Sales Order

Attributes:

Sales Order ID (Primary Key)

Customer ID (Foreign Key)

Product ID (Foreign Key)

Quantity Sold

Order Date

Status

Relationships and Cardinality Ratios

1. Product ↔ Supplier

Relationship: Supplies

Cardinality: Many-to-One (Many products can have one supplier)

2. Product ↔ Stock Level

Relationship: Contains

Cardinality: One-to-One (Each product has one stock level record)

3. Warehouse ↔ Product

Relationship: Stores

Cardinality: Many-to-Many (Products can be stored in multiple warehouses, and each warehouse can hold multiple products)

4. Inventory Transaction ↔ Product

Relationship: Affects

Cardinality: Many-to-One (Many transactions can affect one product)

5. Purchase Order ↔ Supplier

-Relationship: Created by

Cardinality: Many-to-One (Many purchase orders can be issued to one supplier)

6. Purchase Order ↔ Product

Relationship: Contains

Cardinality: Many-to-Many (A purchase order can include multiple products, and a product can be part of multiple orders)

7.Sales Order ↔ Product

Relationship: Contains

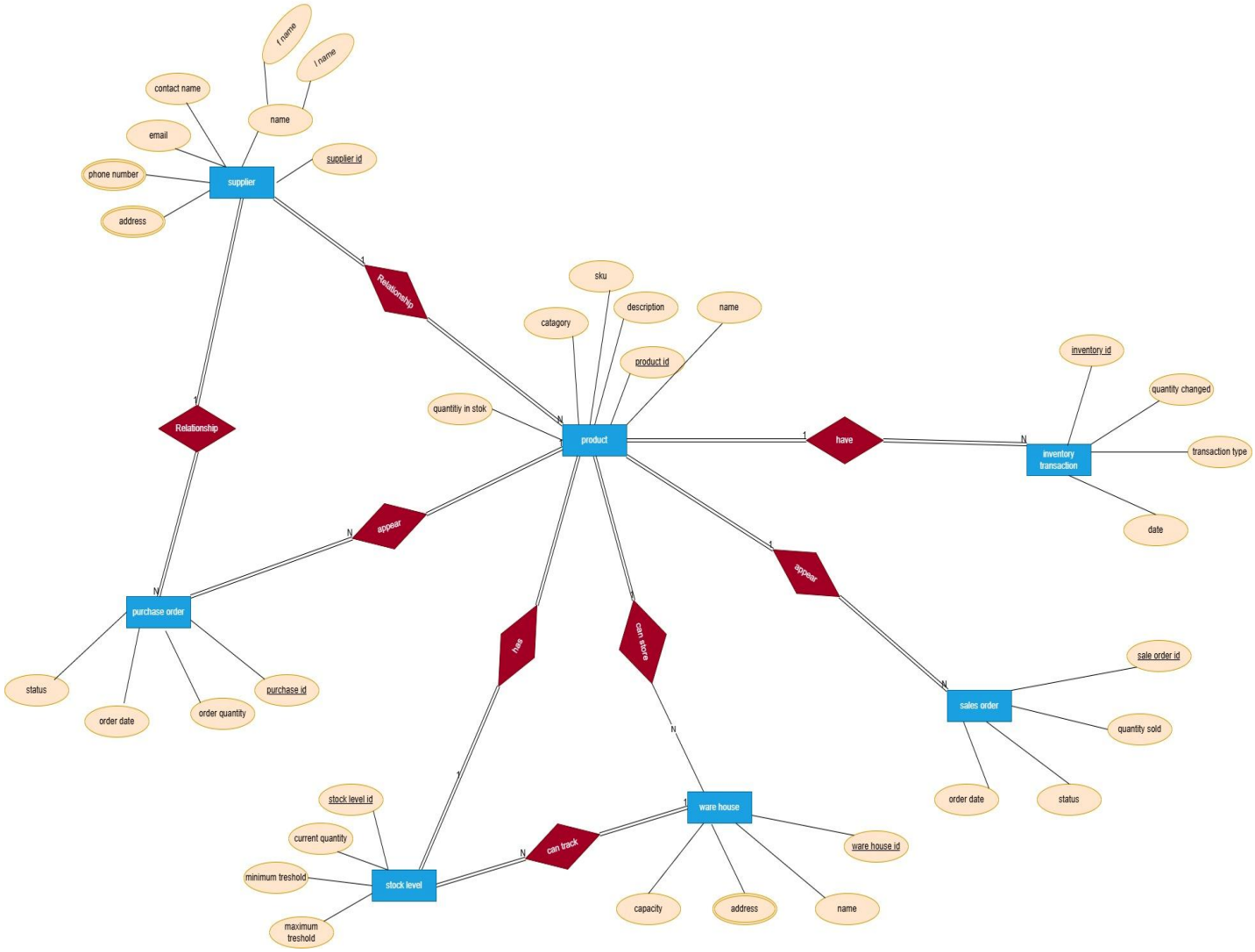
Cardinality: Many-to-Many (A sales order can contain multiple products, and a product can be part of multiple sales orders

CARDINALITY RATIO AND PARTICIPATION FOR ENTITIES

ENTITY	CARDINALITY RATIO	PARTICIPATION
Product & Supplier	Many-to-One	Total

Product & Stock Level	one-to-One	Total
Product & Inventory Transaction	Many-to-One	Total
Product & Purchase Order	Many-to-One	Partial
Product & Sales Order	Many-to-One	Total
Supplier & Product	Many-to-One	Total
Supplier & Purchase Order	Many-to-One	Total
Warehouse& Product	Many-to-One	Total
Warehouse& Stock level	Many-to-One	Total
Stock Level& Product	Many-to-One	Total
Stock Level& Warehouse	one-to-One	Total
Inventory Transaction &Product	Many-to-One	Total
Purchase Order &Supplier	Many-to-One	Total
Purchase Order & Product	Many-to-One	Total
Sales Order& Product	Many-to-One	Total
Sales Order& Customer	Many-to-One	Total

ER DAGRAMDIAGRAM



CONVERT ER DIAGRAM INTO TABLE

PRODUCT

Product id	name	description	Quantity in stock	Suk	Price	category

SUPPILER

Supplier id	name	Contact name	phone	email	address

WAREHOUSE

Warehouse id	name	address	capacity

INVENTORY TRANSACTION

Transaction id	Product id(FK)	Quantity changed	Transaction type	Date

PURCHASE ORDER

Purchase order id	Supplier id	Product id(FK)	Order Quantity	Order date	status

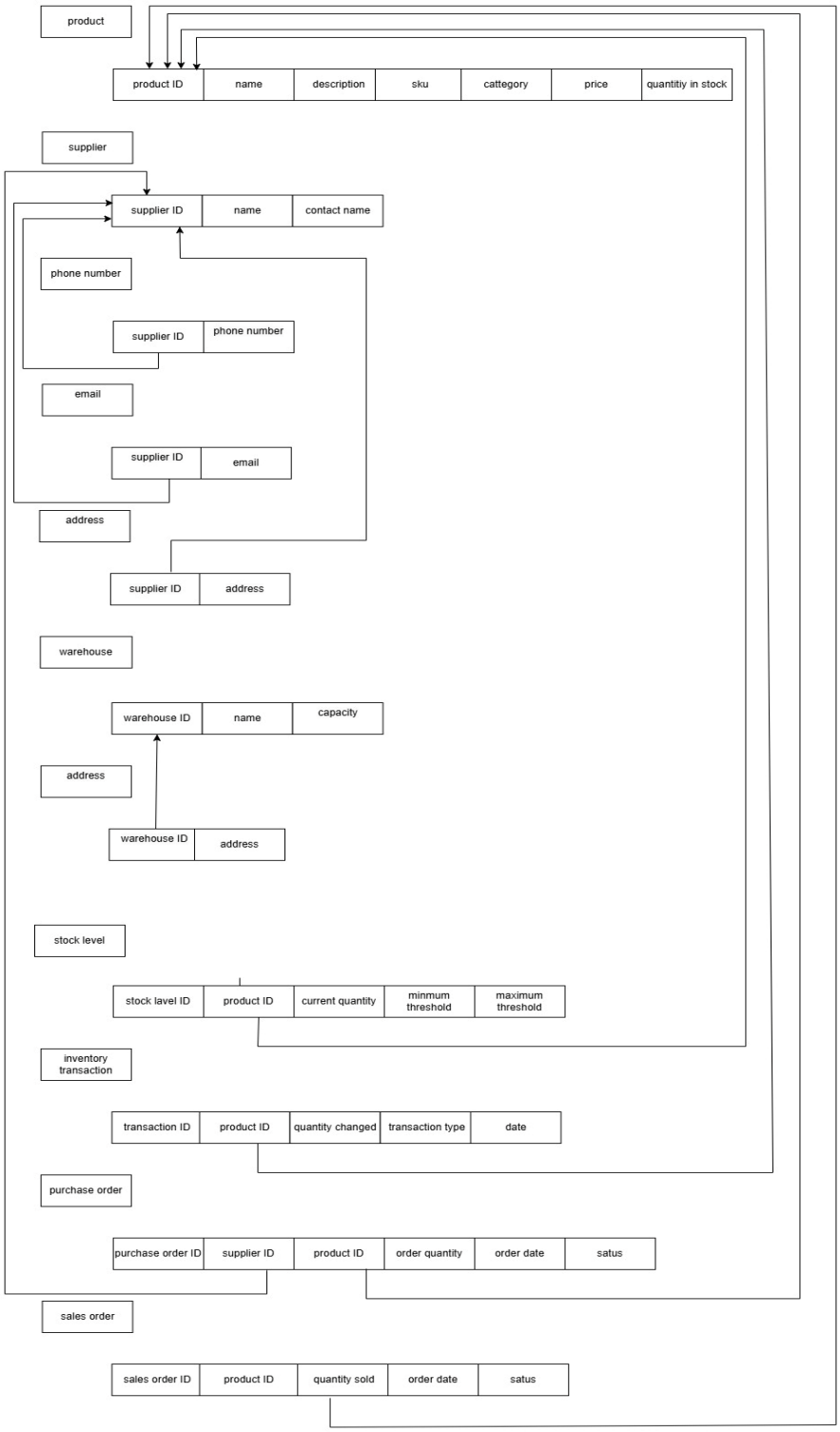
SALES ORDER

Seles order id	Product id(FK)	Quantity sold	Order date	status

STOCK LEVEL

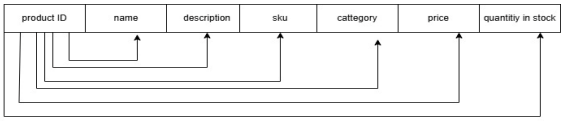
Stock level id	Product id(FK)	Current Quantity	Minimum Threshold	maximum Threshold

MAPPING RELATION

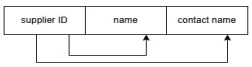


FUNCTIONAL DEPENDENCY

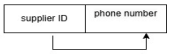
product



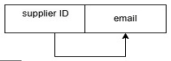
supplier



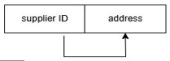
phone number



email

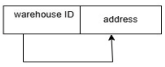


address

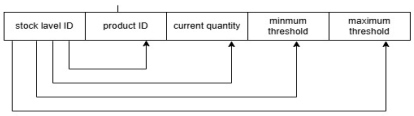


warehouse

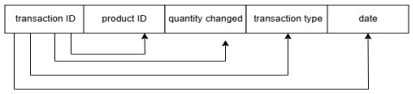
address



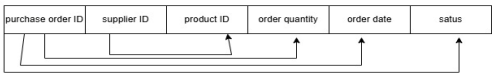
stock level



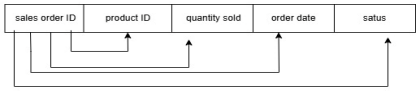
inventory transaction



purchase order



sales order



NORMALIZATION

SUPPILER

Supplier ID	Name	Phone number	Email	Address
1	Tech innovation lcc	0988776566 0711987899	av@thechinnovation.com lcc@innovation.com	AA BD
2	Lap Campany	0955789654 0988776655	Lap@campany.com lapp@campany.com	GG WE

1NF

Supplier ID	Name	Phone number	Email	Address
1	Tech innovation lcc	0988776566	av@thechinnovation.com	AA
1	Tech innovation lcc	0711987899	lcc@innovation.com	BD
2	Lap Campany	0955789654	Lap@campany.com	GG
2	Lap Campany	0988776655	lapp@campany.com	WE

SQL

```sql
CREATE DATABASE INVENTORY
use inventory
CREATE TABLE Product(
ProductID INT PRIMARY KEY,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT
)
CREATE TABLE Supplier (
SupplierID INT PRIMARY KEY,
Name VARCHAR(100),
ContactName VARCHAR(100),
 PhoneNumber INT,
 Email VARCHAR(100),
 Address VARCHAR(255)
)
CREATE TABLE Warehouse (
 WarehouseID INT PRIMARY KEY,
 Name VARCHAR(100),
 Address VARCHAR(255),
 Capacity INT
)
CREATE TABLE StockLevel (
 StockLevelID INT PRIMARY KEY,
 ProductID INT,
 CurrentQuantity INT,
 MinimumThreshold INT,
 MaximumThreshold INT,
 FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
)
CREATE TABLE InventoryTransaction (
 TransactionID INT PRIMARY KEY,
 ProductID INT,
 QuantityChanged INT,
 TransactionType VARCHAR(50),
 Date VARCHAR(55),
 FOREIGn KEY (ProductID) REFERENCES Product(ProductID)
 )
CREATE TABLE PurchaseOrder (
 PurchaseOrderID INT PRIMARY KEY,
 SupplierID INT,
```

```sql
 ProductID INT,
 OrderQuantity INT,
 OrderDate VARCHAR(55),
 Status VARCHAR(50),
 FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID),
 FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
)
CREATE TABLE SalesOrder (
 SalesOrderID INT PRIMARY KEY,
 CustomerID INT,
 ProductID INT,
 OrderQuantity INT,
 OrderDate VARCHAR(77),
 Status VARCHAR(50),
 FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
)
INSERT INTO Product (ProductID, Name, Description, SKU,
Category, Price, QuantityInStock) VALUES
(1, 'Laptop', ' gaming laptop', 'Electronics', 'squ12',44000.00,
10),
(2, 'Printer', 'All-in-one printer', 'Electronics','squ34',
150000.00, 20),
(3, 'Headphones','Noise-cancelling','Electronics','squ56',
20000.00, 25)
INSERT INTO Supplier (SupplierID, Name, ContactName,
PhoneNumber, Email, Address) VALUES
(1, 'hp campany', 'Ali John', 0977665544, 'ali@hpcampes.com', '
AA'),
(2, 'softwer campany.', 'yordanos habtie', 0944778899,
'yor@softcampe.com', 'COM'),
(3, 'undried campany ', 'Dianel smith', 09887766,
'diana@officesupplies.com', 'BD')
INSERT INTO Warehouse (WarehouseID, Name, Address, Capacity)
VALUES
(1, 'Main Warehouse', 'COM', 1000),
(2, 'Secondary Warehouse', 'BD', 8700),
(3, 'Central Warehouse', 'AA', 1100);
INSERT INTO StockLevel (StockLevelID, ProductID,
CurrentQuantity, MinimumThreshold,
MaximumThreshold) VALUES
(5, 1, 30, 6, 40),
(6, 2, 50, 12, 100),
(7, 3, 40, 18, 60);
SELECT* FROM StockLevel;
INSERT INTO InventoryTransaction (TransactionID, ProductID,
QuantityChanged, TransactionType, Date)
VALUES
```

```sql
(5, 1, 5, 'addition', '2022-01-01 '),
(6, 2, -2, 'subtraction', '2024-01-02'),
(7, 3, 8, 'addition', '2025-01-07 ');
INSERT INTO PurchaseOrder(PurchaseOrderID, SupplierID,
ProductID, OrderQuantity, OrderDate, Status)
VALUES
(2,1, 1, 5, '2023-01-05', 'Completed'),
(3, 2, 2, 20, '2022-06-06', 'Pending'),
(4, 3, 3, 12, '2025-01-11', 'Pending');
INSERT INTO SalesOrder (SalesOrderID, CustomerID, ProductID,
OrderQuantity, OrderDate, Status)
VALUES
(3, 5, 1, 11, '2022-01-10', 'Shipped'),
(4, 6, 2, 22, '2024-01-14', 'Processing'),
(5, 7, 3, 33,'2023-01-15', 'Shipped');
BEGIN TRANSACTION;
ALTER TABLE Product ADD Quality VARCHAR(77);
SELECT * FROM Product;
ROLLBACK;

BEGIN TRANSACTION;
ALTER TABLE Product ADD Quality VARCHAR(77);
COMMIT;

BEGIN TRANSACTION;
ROLLBACK;
SELECT * FROM Product;

BEGIN TRANSACTION;
UPDATE Supplier SET Address = 'WDU' WHERE SupplierID = 2;
SELECT*FROM Supplier;
ROLLBACK;

BEGIN TRANSACTION;
UPDATE Supplier SET Address = 'WDU' WHERE SupplierID = 2;
COMMIT;

BEGIN TRANSACTION;
ROLLBACK;
SELECT * FROM Supplier;

BEGIN TRANSACTION;
DELETE FROM Warehouse WHERE WarehouseID = 2;
SELECT * FROM Warehouse;
ROLLBACK;

SELECT * FROM Warehouse;
```

```sql
BEGIN TRANSACTION;
DELETE FROM Warehouse WHERE WarehouseID = 2;
COMMIT;

SELECT * FROM Warehouse;

BEGIN TRANSACTION;
ROLLBACK;
SELECT * FROM Warehouse;

CREATE TRIGGER invn2 ON DATABASE
FOR CREATE_TABLE, ALTER_TABLE, DROP_TABLE
AS
    PRINT 'You cannot create, alter, or drop on this database';
ROLLBACK;

ALTER TABLE Product ADD Address VARCHAR(55);
DROP TRIGGER invn2 ON DATABASE;

CREATE TRIGGER invn ON Product
FOR INSERT, DELETE, UPDATE
AS
    PRINT 'You cannot insert, delete, or update on this table';
ROLLBACK;

INSERT INTO Product (ProductID, Name, Description, SKU,
Category, Price, QuantityInStock)
VALUES (4, 'PHONE', 'gaming', 'squ155', 'Electronics', 42000.00,
19);
DROP TRIGGER invn;

CREATE TRIGGER invn ON Supplier
FOR INSERT, DELETE, UPDATE
AS
    PRINT 'You cannot insert, delete, or update on this table';
ROLLBACK;

UPDATE Supplier SET ContactName = 'PROMI' WHERE SupplierID = 1;
DROP TRIGGER invn;

CREATE TRIGGER invn ON InventoryTransaction
FOR INSERT, DELETE, UPDATE
AS
    PRINT 'You cannot insert, delete, or update on this table';
ROLLBACK;
```

```sql
DELETE FROM InventoryTransaction WHERE TransactionID = 5;
DROP TRIGGER invn;

SELECT * FROM InventoryTransaction;

USE INVENTORY;

CREATE LOGIN admin4 WITH PASSWORD = 'pass1111';
CREATE LOGIN admin5 WITH PASSWORD = 'pass2222';
CREATE LOGIN admin6 WITH PASSWORD = 'pass3333';

CREATE USER kalkien FOR LOGIN admin4;
CREATE USER mogessie FOR LOGIN admin5;
CREATE USER niguse FOR LOGIN admin6;

CREATE ROLE manager;
CREATE ROLE customer;
CREATE ROLE sales;

EXEC sp_addrolemember 'manager', 'kalkien';
EXEC sp_addrolemember 'customer', 'mogessie';
EXEC sp_addrolemember 'sales', 'niguse';

GRANT SELECT, ALTER, INSERT ON Supplier TO  manager WITH GRANT
OPTION;
REVOKE GRANT OPTION FOR INSERT, ALTER ON Supplier FROM  manager
CASCADE;
REVOKE INSERT ON Supplier FROM manager;

GRANT SELECT ON Product TO customer WITH GRANT OPTION;
REVOKE GRANT OPTION FOR SELECT ON Product FROM customer CASCADE;
REVOKE SELECT ON Product FROM customer;

GRANT ALTER, INSERT ON InventoryTransaction TO sales WITH GRANT
OPTION;
REVOKE GRANT OPTION FOR ALTER, INSERT ON InventoryTransaction
FROM sales CASCADE;
REVOKE ALTER ON InventoryTransaction FROM sales;

BACKUP DATABASE INVENTORY
TO DISK ='C:\KK\Inventory_Backup.bak'


DROP DATABASE INVENTORY

RESTORE DATABASE INVENTORY
FROM DISK = 'C:\KK\Inventory_Backup.bak'
```

```sql
CREATE DATABASE INVENTORY2020
USE INVENTORY2020
CREATE TABLE Product1(
ProductID INT PRIMARY KEY,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT,num1 int not null
)
select * from Product1
INSERT INTO Product1  VALUES(1, 'Laptop', ' gaming laptop',
'Electronics', 'squ12',44000.00, 10,1)
INSERT INTO Product1  VALUES(2, 'Printer', 'All-in-one printer',
'Electronics','squ34', 150000.00, 20,3)
INSERT INTO Product1  VALUES(3, 'Headphones','Noise-
cancelling','Electronics','squ56', 20000.00, 25,5)

CREATE TABLE InventoryTransaction4 (
 TransactionID INT PRIMARY KEY,
 ProductID INT,
 QuantityChanged INT,
 TransactionType VARCHAR(50),
 Date VARCHAR(55),
 FOREIGn KEY (ProductID) REFERENCES Product1(ProductID),num2 int
not null)

 INSERT INTO InventoryTransaction4 (TransactionID, ProductID,
QuantityChanged, TransactionType, Date, num2)
VALUES
(5, 1, 5, 'addition', '2022-01-01 ',1),
(6, 2, -2, 'subtraction', '2024-01-02',6),
(7, 3, 8, 'addition', '2025-01-07 ',3);

select * from InventoryTransaction4
select * from Product1 inner join InventoryTransaction4 on
Product1.num1=InventoryTransaction4.num2
select * from Product1 full outer join InventoryTransaction4 on
Product1.num1=InventoryTransaction4.num2
select * from Product1 left outer join InventoryTransaction4 on
Product1.num1=InventoryTransaction4.num2
select * from Product1 right outer join InventoryTransaction4 on
Product1.num1=InventoryTransaction4.num2
  create database INVENTORY
 use INVENTORY
 CREATE TABLE Product(
```

```
ProductID INT PRIMARY KEY,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT
);
INSERT INTO Product (ProductID, Name, Description, SKU,
Category, Price, QuantityInStock) VALUES
(1, 'Laptop', ' gaming laptop', 'Electronics', 'squ12',44000.00,
10),
(2, 'Printer', 'All-in-one printer', 'Electronics','squ34',
150000.00, 20),
(3, 'Headphones','Noise-cancelling','Electronics','squ56',
20000.00, 25)

 select*from Product

     --VERTICAL FRAGMENTS OF DATA
      --fragment 1
 select  ProductID, Name, Description, SKU into Product_A from
Product;
 select *from Product_A

 --fragment 2
 select  ProductID ,Category, Price, QuantityInStock into
Product_B from Product
 select *from Product_B
       --RECONSTRUCT THE ORIGINAL TABLE(JOIN)
 select X.ProductID, X.Name,X.Description, X.SKU ,
 Y.Category, Y.Price, Y.QuantityInStock from Product_A X join
Product_B Y on X.ProductID=Y.ProductID;

              -- HORIZONTAL FRAGMENTS OF DATA
               --fragment 1
               select *  into Product_D from Product where
ProductID=1;
               select * from Product_D

               --fragment 2

               select *  into Product_E1 from Product where
Name='Headphones';
                    select * from Product_E1
                    --fragment 3
```

```
                    select *  into Product_A1 from Product where
Name='Printer';
                    select * from Product_A1

                --RECONSTRUCT THE ORIGINAL TABLE WITH union
                select* from Product_D union select* from
Product_E1 union select * from Product_A1


--DATA REPLICATION
          create database supe1
        use supe1
CREATE TABLE Product(
ProductID INT,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT
);
select* from Product

CREATE TABLE Product1(
ProductID INT ,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT
)
select* from Product1

        create database sub2

         use  sub2
CREATE TABLE Product(
ProductID INT ,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT
)
```

```sql
INSERT INTO Product (ProductID, Name,
 Description, SKU, Category, Price,
 QuantityInStock) VALUES
(1, 'Laptop', ' gaming laptop', 'Electronics', 'squ12',44000.00,
10),
(2, 'Printer', 'All-in-one printer', 'Electronics','squ34',
150000.00, 20),
(3, 'Headphones','Noise-cancelling','Electronics','squ56',
20000.00, 25)

select* from Product

insert into supe1link.supe1.dbo.Product( ProductID, Name,
 Description, SKU, Category, Price,
 QuantityInStock )
  select ProductID, Name,
 Description, SKU, Category, Price,
 QuantityInStock from  sub2.dbo.Product;


 CREATE TABLE Product1(
ProductID INT ,
Name VARCHAR(100),
Description VARCHAR(330),
SKU VARCHAR(50),
Category VARCHAR(50),
Price DECIMAL(10, 2),
QuantityInStock INT
)
INSERT INTO Product1(ProductID, Name,
 Description, SKU, Category, Price,
 QuantityInStock) VALUES
(1, 'desktop', ' gaming laptop', 'Electronics',
'squ12',66666.00, 17),
(2, 'swich', 'All-in-one printer', 'Electronics','squ34',
1000.00, 18),
(3, 'pc','Noise-cancelling','Electronics','squ56', 60000.00, 22)

select* from Product1

insert into supe1link.supe1.dbo.Product( ProductID, Name,
 Description, SKU, Category, Price,
 QuantityInStock )
   select ProductID, Name,
 Description, SKU, Category, Price,
 QuantityInStock from  sub2.dbo.Product;
```

```sql
select * from Product1;
     use master

     EXEC sp_addlinkedserver
     @server='supe1link',
     @srvproduct='',
     @provider='SQLNCLI11',
     @datasrc='localhost\SQLEXPRESS';

     EXEC sp_addlinkedsrvlogin
     @rmtsrvname='supe1link',
     @useself='true';
```

Conclusion

An Inventory Control Management System is essential for modern businesses to maintain effective inventory levels and streamline operations. By leveraging technology, companies can enhance efficiency, reduce costs, and improve customer satisfaction, ultimately driving growth and profitability. Implementing an ICMS leads to better decision-making through data-driven insights and helps businesses adapt to changing market demands.

An Inventory Control Management System is an essential tool for businesses looking to enhance their inventory management processes. By offering features that streamline tracking, ordering, and reporting, an ICMS not only improves operational efficiency but also helps in achieving better financial outcomes. As businesses face increasing competition and market demands, implementing an effective ICMS becomes crucial for sustained growth and success.

An ICMS is vital for businesses aiming to enhance their inventory practices. By improving tracking, ordering, and reporting, it fosters operational efficiency and supports financial performance, making it crucial for competitiveness in today's market.